UNIVERSITE CATHOLIQUE D'AFRIQUE CENTRALE
INSTITUT CATHOLIQUE DE YAOUNDE
Faculté de Sciences Sociales et de Gestion

A·O·J·F·A

EXAMINATION OF FINANCIAL MATHEMATICS

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First Semester 2022-2023

BEM 3

NB: All final answers at 2 decimal places and all final answers in a box

Exercise 1 (6 Marks)

1) on the 30th December 2023, Mary has to be credited a sum of 2200000frs. On the 30th December 2006 she sells this credit title and the settlement was done in cash at an annual rate of 7.5 %. She receives this amount and immediately places it in a bank at an annual rate of 11.5 % up till the 30th December 2023

a) what will be her capital on the 30th December 2006

b) what will be her capital on the 30th December 2023

c) At what date will she have a sum of 1650000frs

d) what amount could be deducted from the sum received on the 30th December 2006, so that the balance placed under the same conditions would earn her 2200000frs

Exercise 2 (10 Marks)

2) With the objective of creating a business after university education, a student at BME1 decides to sacrifice a constant sum of 8500frs every month to be saved at a bank at a monthly rate of 8.5% from September 2019 to June 2023.

a) what will be the value of this capital on August 2019

b) what will be the value of this capital on June 2023

c) what will be the value of this capital five month after the last placement

if the student inherits a sum of 1500000frs in march 2022, which is immediately kept in the bank in the same period at a rate of 10 %

d) with this additional information what will he receive in June 2023 taking into account his inheritance.

If the total capital plus interest generated in d) above was loaned to a friend to be reimbursed in six years, at an annual rate of 11.5%, draw the amortization table for the following reimbursement modalities

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e) First modality: Reimbursement in fine.

f) Second modality: Reimbursement by constant amortization.

Exercise 3 (4 Marks)

3) a) An individual saves in a financial establishment a sum of 145000frs, which last with the bank for 10 semesters at an annual rate of 10.5% per year, determine the interest and acquired value

b) given a capital of 25000frs placed over a duration of 426 days had a acquired value of 87000frs, determine the interest rate

c) Given an original capital of 40000frs, capitalized semestrial at a semestrial rate of 4.75%, gave an acquired value of 76597.84frs. Determine the duration of the placement in semester.

Good Luck